

# Inflation Dynamics - Computational Notebook

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## Abstract

This book presents computation details of estimating inflation dynamics.

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This notebook presents computation details of estimating inflation dynamics.

So far we have covered estimation of

1. GARCH
2. EGARCH
3. GJR-GARCH
4. SPARCH
5. Regime-switching GARCH
6. BEGE GARCH

State variable definitions:

1.  $\pi_t$ : inflation.
2.  $\mathcal{F}_{t-1}$ : filtration at time  $t - 1$ .
3.  $SPF_{t-1}$ : professional forecast on  $\pi_t$  inflation measured by  $\mathcal{F}_{t-1}$ .
4.  $\hat{\pi}_t$ : expected inflation  $E[\pi_t \mid \mathcal{F}_{t-1}]$ .
5.  $u_t$ : inflation innovation (residual), defined as  $u_t := \pi_t - \hat{\pi}_t$ .
6.  $u_t^+$ : positive residual, defined as  $u_t^+ := u_t \cdot \mathbf{1}(u_t > 0)$
7.  $u_t^-$ : negative residual, defined as  $u_t^- := u_t \cdot \mathbf{1}(u_t \leq 0)$

## 1. INFLATION SUMMARY STATISTICS

I start with simple statistical summary by plotting inflation and professional forecast over time.

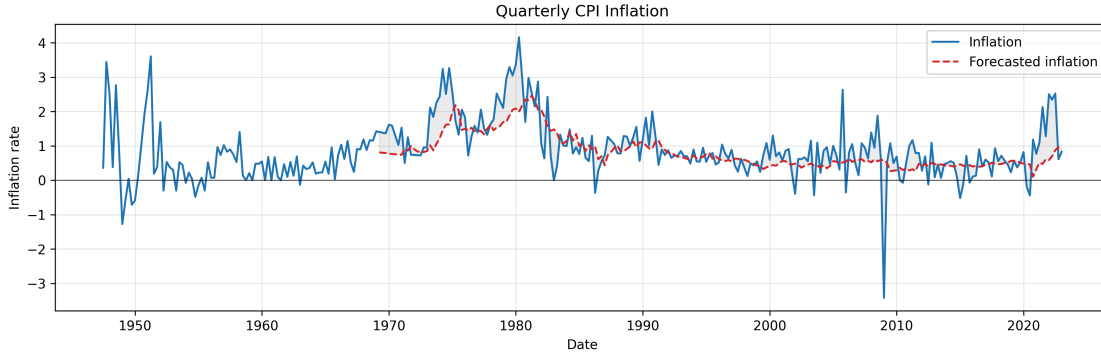


Figure 1: Quarterly inflation  $\pi_t$  and profesional forecast  $SPF_{t-1}$ .

Notice that inflation was recorded from 1947 but profesional forecast started from 1969.

|               | Inflation | Forecasted inflation |
|---------------|-----------|----------------------|
| Date Start    | 1947Q2    | 1969Q1               |
| Date End      | 2022Q4    | 2022Q4               |
| #Observations | 303       | 216                  |

Table 1: Inflation statistics

### 1.a. Effective Sample

To make log-likelihood values comparable across specifications, I use an effective sample for all mean models and all volatility models.

In ARX models, lagged regressors (for example,  $SPF_{t-1}$ ) are not observed at the very beginning of the raw sample. Therefore, I start the estimation sample at the first date where both current forecast inflation ( $SPF_t$ ), and one-lag forecast inflation ( $SPF_{t-1}$ ) are available. For the quarterly data, forecast inflation ( $SPF_t$ ) is available from **1969Q1** to **2022Q4** (216 observations), requiring one lag ( $SPF_{t-1}$ ) shifts the usable start to **1969Q2**. So the trimmed estimation window is **1969Q2–2022Q4**, with **215 observations**. This same trimmed sample is then used for all mean specifications, so the residual series has the same length across models.

For GARCH-type recursion, the first few conditional variances are unobserved (for example,  $\sigma_0^2, \sigma_{-1}^2, u_0^2, u_{-1}^2$ ). I initialize these pre-sample variances using the model-implied unconditional variance, for example in GARCH model:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \sum_i \alpha_i - \sum_j \beta_j}, \quad (1)$$

and set

$$u_0^2 = u_{-1}^2 = \dots = \sigma_0^2 = \sigma_{-1}^2 = \dots = \bar{\sigma}^2. \quad (2)$$

### 1.b. Effective Sample Summary Statistics

I report the summary statistics for trimmed effect sample. Skewness and Kurtosis are adjusted by sample size.

|               | Inflation | SPF    | SPF_shock |
|---------------|-----------|--------|-----------|
| Date Start    | 1969Q2    | 1969Q2 | 1969Q2    |
| Date End      | 2022Q4    | 2022Q4 | 2022Q4    |
| #Observations | 215       | 215    | 215       |
| Mean          | 0.9918    | 0.8320 | 0.1598    |
| Median        | 0.7937    | 0.6296 | 0.1101    |
| Std           | 0.8580    | 0.4969 | 0.6642    |
| Min           | -3.4170   | 0.1048 | -4.0117   |
| Max           | 4.1612    | 2.4566 | 2.1729    |
| P5            | -0.0681   | 0.3482 | -0.7745   |
| P25           | 0.5328    | 0.4748 | -0.1669   |
| P75           | 1.2913    | 1.0150 | 0.4617    |
| P95           | 2.5883    | 2.0030 | 1.2581    |
| Skewness      | 0.3550    | 1.3633 | -0.7131   |
| Kurtosis      | 3.8995    | 1.2053 | 7.3006    |
| AC(1)         | 0.5981    | 0.9717 | 0.2764    |
| AC(2)         | 0.5353    | 0.9473 | 0.1322    |
| AC(4)         | 0.4708    | 0.8929 | 0.0881    |
| AC(12)        | 0.2918    | 0.6949 | 0.0011    |

Table 2: Inflation statistics

## 2. MEAN PROCESS SPECIFICATION

Four types of mean processes are of interest:

- **Constant:**  $\pi_{t+1} = SPF_t + \mu_{t+1}$
- **ARX(1,1):**  $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,1):**  $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,2)<sup>1</sup>:**  $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + \mu_{t+1}$

where  $\mu_{t+1}$  is the residual term.

### 2.a. OLS Results

Estimation results of four mean models with HAC standard errors in parentheses below each estimate. (The log-likelihood is computed under the Gaussian assumption.)

#### Constant:

$$\hat{\pi}_{t+1} = SPF_t \quad (3)$$

No estimated coefficients. Residuals  $\mu_{t+1} = \pi_{t+1} - SPF_t$ .

- Log-likelihood: **-222.664**
- AIC: **447.329**, BIC: **450.700**
- Residual skewness: **-0.713**, excess kurtosis: **7.301**

#### ARX(1,1):

$$\hat{\pi}_{t+1} = \begin{matrix} 0.0824 & +0.3005 \pi_t & +0.7337 SPF_t \\ (0.086) & (0.112) & (0.167) \end{matrix} \quad (4)$$

- Log-likelihood: **-207.381**
- AIC: **420.762**, BIC: **430.874**
- Residual skewness: **-1.047**, excess kurtosis: **8.210**

#### ARX(2,1):

$$\hat{\pi}_{t+1} = \begin{matrix} 0.0897 & +0.2892 \pi_t & +0.0834 \pi_{t-1} & +0.6385 SPF_t \\ (0.086) & (0.098) & (0.126) & (0.251) \end{matrix} \quad (5)$$

- Log-likelihood: **-206.810**
- AIC: **421.619**, BIC: **435.102**
- Residual skewness: **-1.191**, excess kurtosis: **9.050**

#### ARX(2,2):

$$\hat{\pi}_{t+1} = \begin{matrix} 0.0856 & +0.2992 \pi_t & +0.0914 \pi_{t-1} & +0.4084 SPF_t & +0.2136 SPF_{t-1} \\ (0.086) & (0.106) & (0.125) & (0.433) & (0.297) \end{matrix} \quad (6)$$

- Log-likelihood: **-206.660**
- AIC: **423.319**, BIC: **440.172**
- Residual skewness: **-1.212**, excess kurtosis: **9.129**

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<sup>1</sup>In the GARCH family estimation exercise, we found that ARX(2,2) is always dominated by other three specifications. Thus, we stop estimating ARX(2,2) in BEGE estimation to saving computational resources.

### 3. GARCH FAMILY

This section feature GARCH, GJR-GARCH and EGARCH with three residual distributions–normal, student-t and mixture of normal.

#### 3.a. Volatility Recursion Specification

**GARCH(p,q):**

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i u_{t-i}^2 + \sum_{k=1}^q \beta_k \sigma_{t-k}^2 \quad (7)$$

**GJR-GARCH(p,q):**

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i |u_{t-i}^+|^2 + \sum_{j=1}^p \gamma_j |u_{t-j}^-|^2 + \sum_{k=1}^q \beta_k \sigma_{t-k}^2 \quad (8)$$

**EGARCH(p,q):**

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i (|e_{t-i}| - \sqrt{2/\pi}) + \sum_{j=1}^o \gamma_j e_{t-j} + \sum_{k=1}^q \beta_k \ln \sigma_{t-k}^2 \quad (9)$$

where  $e_{t-j} = u_{t-j}/\sigma_{t-j}$ .

#### 3.b. Log Likelihood Specification

##### 3.b.i. Normal distribution:

This package simultaneously estimates mean and variance parameters by minimizing Log Likelihood, which is calculated by Iteration:

1. Compute residuals  $\varepsilon_t$  of mean model
2. Compute conditional variance  $h_t$  using VolatilityModel
3. Compute likelihood function by

$$\ln L = \sum_{t=1}^T -\frac{1}{2} \left( \ln 2\pi + \ln h_t + \frac{\varepsilon_t^2}{h_t} \right) \quad (10)$$

where  $\varepsilon_t$  is residual and  $h_t$  is conditional variance at time t, Using scipy.optimizer and SLSQP kernel.

$$L = \prod_{t=1}^T \frac{1}{\sqrt{2\pi h_t}} \exp \left\{ -\frac{u_t^2}{2h_t} \right\} \quad (11)$$

and

$$\ln L = \sum_{t=1}^T \left( -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln h_t - \frac{1}{2} \frac{u_t^2}{h_t} \right) \quad (12)$$

##### 3.b.ii. Student t distribution:

One parameter :  $\nu$

Log Likelihood Function of one residual  $\varepsilon_t$  with conditional standard error  $\sigma_t$  :

$$\ln \Gamma\left(\frac{\nu+1}{2}\right) - \ln \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \ln(\pi(\nu-2)\sigma_t^2) - \frac{\nu+1}{2} \ln(1 + \varepsilon_t^2/(\sigma_t^2(\nu-2))) \quad (13)$$

where  $\Gamma$  is the gamma function.

### 3.b.iii. Gaussian Mixture distribution:

The mixture of 2 normal distribution given parameters  $p_1, \mu_1, \sigma_1^2$  is

$$f(z_t) = p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{(z_t - \mu_1)^2}{2\sigma_1^2}\right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left\{-\frac{(z_t - \mu_2)^2}{2\sigma_2^2}\right\} \quad (14)$$

where  $p_2 = 1 - p_1$ ,  $\mu_2 = \frac{-p_1\mu_1}{p_2}$ , and  $\sigma_2^2 = \frac{1-p_2\mu_2^2-p_1(\sigma_1^2+\mu_1^2)}{p_2}$

N th order moment can be calculated as:

$$E[z^n] = \int z^n \left[ p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{(z - \mu_1)^2}{2\sigma_1^2}\right\} + (1 - p_1) \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left\{-\frac{(z - \mu_2)^2}{2\sigma_2^2}\right\} \right] dz \quad (15)$$

$$E[z^n] = p_1 E[x_1^n | x_1 \sim N(\mu_1, \sigma_1^2)] + p_2 E[x_2^n | x_2 \sim N(\mu_2, \sigma_2^2)] \quad (16)$$

Similary, CDF can be calculated analytically by

$$\Phi(a) = \int_{-\infty}^a z \left[ p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{(z - \mu_1)^2}{2\sigma_1^2}\right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left\{-\frac{(z - \mu_2)^2}{2\sigma_2^2}\right\} \right] dz \quad (17)$$

$$\Phi(a) = p_1 \Phi_{x_1}(a) + p_2 \Phi_{x_2}(a) \quad (18)$$

We can think mixture of two normal distributions generated by three random variable  $Z, X_1, X_2$ .  $Z$  follows Bernoulli( $P_1$ ),  $X_1 \sim N(\mu_1, \sigma_1^2)$  and  $X_2 \sim N(\mu_2, \sigma_2^2)$ . In order to generate a random sample point, we can first use  $Z$  to decide which normal distributions to use, and then randomly pick a point in that normal distribution.

Log likelihood function is

$$L(u_t) = \sum_{t=1}^T \ln \frac{1}{\sigma_t} \left[ p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{(z_t - \mu_1)^2}{2\sigma_1^2}\right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left\{-\frac{(z_t - \mu_2)^2}{2\sigma_2^2}\right\} \right] \quad (19)$$

, where  $z_t = \frac{u_t}{\sigma_t}$

### 3.c. Estimation Summary

### 3.d. Best Estimation Illustrations

**Table 2: Model selection by AIC: GJR vs EGARCH**

| Distribution | GJR Mean | GJR Vol | GJR AIC  | EGARCH Mean | EGARCH Vol | EGARCH AIC |
|--------------|----------|---------|----------|-------------|------------|------------|
| Normal       | (1,1)    | (2,1)   | 359.4710 | (1,1)       | (1,1)      | 368.2429   |
| $t$          | (1,1)    | (2,1)   | 343.1431 | (2,1)       | (1,1)      | 344.6254   |
| Skew $t$     | (1,1)    | (2,1)   | 345.1312 | (2,1)       | (1,1)      | 346.5791   |
| GED          | (1,1)    | (2,1)   | 345.1925 | (1,1)       | (1,1)      | 347.6934   |

| Distribution  | GJR Mean | GJR Vol | GJR AIC  | EGARCH Mean | EGARCH Vol | EGARCH AIC |
|---------------|----------|---------|----------|-------------|------------|------------|
| Mix of Normal | (1,1)    | (2,1)   | 345.8222 | (2,1)       | (2,1)      | 343.5023   |

**Table 3: Model selection by BIC: GJR vs EGARCH**

| Distribution  | GJR Mean | GJR Vol | GJR BIC  | EGARCH Mean | EGARCH Vol | EGARCH BIC |
|---------------|----------|---------|----------|-------------|------------|------------|
| Normal        | FC       | (2,1)   | 381.4280 | FC          | (1,1)      | 383.4754   |
| $t$           | FC       | (1,1)   | 367.8350 | FC          | (1,1)      | 367.1732   |
| Skew $t$      | FC       | (1,1)   | 370.4242 | FC          | (1,1)      | 369.6649   |
| GED           | FC       | (1,1)   | 371.8489 | FC          | (1,1)      | 370.9644   |
| Mix of Normal | FC       | (1,1)   | 373.7106 | FC          | (1,1)      | 371.0689   |

#### 4. BEGE DENSITY

BEGE density function  $f_{BEGE}(\mu | p, n, \sigma_p, \sigma_n)$  is the function that calculates the density of the observation  $\mu$  given the parameters  $\{p, n, \sigma_p, \sigma_n\}$ .

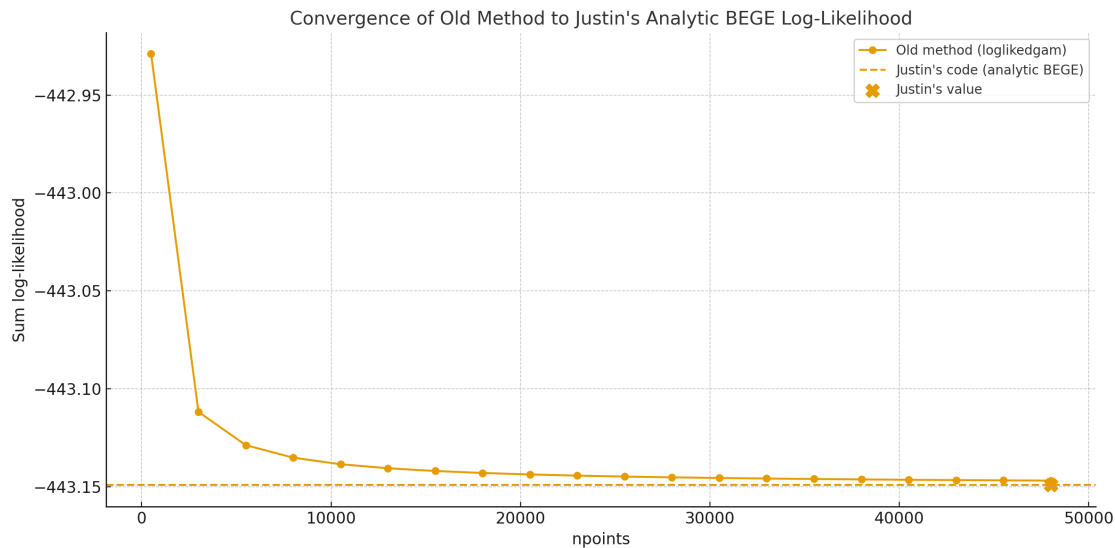
To compare Justin's analytic BEGE density code with the old numerical one, I conducted an experiment using synthetic data. I generated 200 independent observations from a standard normal distribution and fixed the BEGE parameters to

$$p = 5, \quad n = 7, \quad \sigma_p = 0.8, \quad \sigma_n = 1.2. \quad (20)$$

I computed the sum of log-likelihood using two approaches:

1. The *old BEGE function*, which approximates the density via numerical integration on a uniform grid, and whose accuracy depends strongly on the number of grid points; and
2. *Justin's analytic BEGE function*.

The old numerical method was evaluated over a wide range of grid resolutions (up to 50,000 points), and the resulting log-likelihood sums were plotted alongside the benchmark value computed from Justin's code.



#### Findings:

- The old numerical likelihood converges toward Justin's analytic likelihood as the number of grid points increases, but the convergence is slow and requires very fine grids.
- For small or moderate grid sizes, the numerical integration *overestimates* the log-likelihood. This explains why, for a given set of parameters, Justin's implementation produces a lower log-likelihood than the old code.
- Justin's analytic BEGE formula provides a more accurate and more computationally efficient evaluation.

#### 5. BEGE GARCH FAMILY



To start the random search of initial mean parameters, I draw uniform samples from  $(\mu - 2\sigma, \mu + 2\sigma)$  where  $\mu$  and  $\sigma$  are mean and standard deviation from OLS regression. I also set the AR coefficient bound to avoid the AR process to explode. We have four types of mean processes.

**Table 1: Parameter Bounds for Mean Process Specifications**

| Model    | $c$                        | $\rho_1$          | $\rho_2$          | $\phi_1$    | $\phi_2$    |
|----------|----------------------------|-------------------|-------------------|-------------|-------------|
| Constant | —                          | —                 | —                 | —           | —           |
| ARX(1,1) | $(\min \pi_t, \max \pi_t)$ | $(-0.999, 0.999)$ | —                 | $(-10, 10)$ | —           |
| ARX(2,1) | $(\min \pi_t, \max \pi_t)$ | $(-1.999, 1.999)$ | $(-0.999, 0.999)$ | $(-10, 10)$ | —           |
| ARX(2,2) | $(\min \pi_t, \max \pi_t)$ | $(-1.999, 1.999)$ | $(-0.999, 0.999)$ | $(-10, 10)$ | $(-10, 10)$ |

#### 5.a. Constant BEGE

The first model that I estimated is BEGE with invariant shape parameters, which is

$$\begin{aligned}\pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1} \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n\end{aligned}\tag{21}$$

where

$$\omega_p \sim \tilde{\Gamma}(\bar{p}, 1), \quad \omega_n \sim \tilde{\Gamma}(\bar{n}, 1).\tag{22}$$

$\bar{p}$  and  $\bar{n}$  are unconditional shape parameters. The random search and bounds are set to be:

- $0.05 < \sigma_p, \sigma_n < 2$
- $0.1 < p, n < 10$

#### 5.b. Symmetric BEGE

The model assumes different scaled parameters and different constants in GJR recursions for both components:

$$\begin{aligned}\pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n,\end{aligned}\tag{23}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1).\tag{24}$$

Here  $n_t, p_t$  is generated recursively through a GJR-type updating equation with parameters  $(p_0, n_0, \rho, \phi^+, \phi^-)$ :

$$\begin{aligned}p_t &= p_0 + \rho p_{t-1} + \frac{\phi^+}{2\sigma_p^2} (u_{t-1}^+)^2 + \frac{\phi^-}{2\sigma_p^2} (u_{t-1}^-)^2, \\ n_t &= n_0 + \rho n_{t-1} + \frac{\phi^+}{2\sigma_n^2} (u_{t-1}^+)^2 + \frac{\phi^-}{2\sigma_n^2} (u_{t-1}^-)^2\end{aligned}\tag{25}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$ ,
- $10^{-5} < \rho < 0.999$ ,
- $10^{-5} < \phi^+, \phi^- < 0.999$ ,

- $10^{-5} < \sigma_p, \sigma_n < 2$ .

### 5.c. Bad and Good Symmetric GARCH Model

Here  $n_t, p_t$  is generated recursively with parameters  $(p_0, n_0, \rho_p, \rho_n, \phi_p, \phi_n)$ :

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p}{2\sigma_p^2} (u_{t-1})^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n}{2\sigma_n^2} (u_{t-1})^2 \end{aligned} \quad (26)$$

I have set the constraints below.

- $\rho + \phi < 1$  for both good and bad shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$ .
- $\max\{p_t, n_t\} < 200$

### 5.d. Inflation and Deflation Model

Here  $n_t, p_t$  is generated recursively with parameters  $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_n^-)$ :

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} (u_{t-1}^+)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2} (u_{t-1}^-)^2 \end{aligned} \quad (27)$$

According to the emails, I have set the constraints below.

- $\rho_p + \frac{\phi_p^+}{2} < 1$  and  $\rho_n + \frac{\phi_n^-}{2} < 1$ .
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$ .
- $\max\{p_t, n_t\} < 200$

### 5.e. Full BEGE

BEGE GARCH has mean process and higher:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n, \end{aligned} \quad (28)$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \quad (29)$$

Here  $n_t, p_t$  is generated recursively through a GJR-type updating equation with parameters  $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^-)$ :

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} (u_{t-1}^+)^2 + \frac{\phi_p^-}{2\sigma_p^2} (u_{t-1}^-)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2} (u_{t-1}^+)^2 + \frac{\phi_n^-}{2\sigma_n^2} (u_{t-1}^-)^2 \end{aligned} \quad (30)$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$ ,
- $10^{-5} < \rho_p, \rho_n < 0.999$ ,

- $10^{-5} < \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^- < 0.999$ ,
- $10^{-5} < \sigma_p, \sigma_n < 2$ .

According to the emails, I have set the constraints below.

- $\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1$  for both BE and GE shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$  where  $\text{Var}(\pi_t) = 0.75$  is the unconditional variance of inflation.
- $\max\{p_t, n_t\} < 200$ .

## 6. RESULT SUMMARY

Overall, most of the estimations with random initial values converge. I randomly sampled 100,000 starting values for the constant BEGE model, and sampled 10,000 for symmetric BEGE because it is more time consuming. I report the best estimation below.